

III SEMESTER

DIPLOMA IN COMPUTER ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
CURRICULUM-2023 (III Semester)

Sub Code	Name of the Subject	Instruction Periods/Week		Total Periods Per Semester	Scheme Of Examinations			
		Theory	Practicals		Duration (hrs)	Sessional Marks	End Exam Marks	Total Marks
THEORY SUBJECTS								
CM-301	Mathematics – II	4		60	3	20	80	100
CM-302	Digital Electronics	5	-	75	3	20	80	100
CM-303	Operating systems	4	-	60	3	20	80	100
CM-304	Data Structures through C	5	-	75	3	20	80	100
CM-305	DBMS	5	-	75	3	20	80	100
PRACTICAL SUBJECTS								
CM-306	Digital Electronics Lab	-	3	45	3	40	60	100
CM-307	Data Structures Through C Lab	-	6	90	3	40	60	100
CM-308	DBMS Lab	-	4	60	3	40	60	100
CM-309	Multimedia Lab	-	3	45	3	40	60	100
	ACTIVITIES	-	3	45				
	Total	23	19	630		260	640	900

301 common with all branches

302 common with DIOT

303 common with DAIML,DCCN,DIOT

304 common with DCAI, DCBD,DWD

305 common with DAIML,DCAI,DCCN,DWD,DIOT

307 common with DCAI, DCBD,DWD&308 common with DAIML,DCAI,DCCN,DW&309 common with DCCN

C-23 CM-301
ENGINEERING MATHEMATICS-II
(Common to CM/AIML/AMG/AMT/CAI/CCB/CCN/WD)

Course Code	Course Title	No. of Periods/week	Total No. of periods	Marks for FA	Marks for SA
CM - 301	Engineering Mathematics-II	4	60	20	80

Chapter. No	Unit Title	No. of periods	COs mapped
1	Integral Calculus	22	CO1
2	Differential Equations	8	CO2
3	Graph Theory and Probability	17	CO3
4	Statistics	13	CO4
	Total Periods	60	

Course Objectives	(iii) To understand the concepts of indefinite integration and definite integration. (iv) To understand the formation of differential equations and learn various methods of solving first order differential equations. (v) To comprehend the concepts of graph theory and probability. (vi) To learn different statistical techniques for data analysis.
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Course Outcomes	CO1	Integrate various functions using different methods and evaluate definite integrals.
	CO2	Obtain differential equations and solve differential equations of first order and first degree.
	CO3	Able to define the basic concepts of Graph Theory and use the principles of Probability in computational systems.
	CO4	Apply various statistical techniques for data analysis.

C-23 WD-301
ENGINEERING MATHEMATICS – II
(Common to CM/AIML/AMG/AMT/CAI/CCB/CCN/WD)
Learning Outcomes

Unit-I
Integral Calculus

C.O. 1 Integrate various functions using different methods and evaluate definite integrals.

L.O.1.1. Explain the concept of Indefinite integral as an anti-derivative.

- 1.2. State the indefinite integral of standard functions and properties of $\int (u + v) dx$ and $\int k u dx$ where u, v are functions of x and k is constant.
- 1.3. Solve problems involving standard functions using these properties.
- 1.4. Evaluate integrals involving simple functions of the following type by the method of substitution.

i) $\int f(ax + b) dx$, where $f(x)$ is in standard form.

ii) $\int (f(x))^n f'(x) dx$, $n \neq -1$

iii) $\int \frac{1}{f(x)} dx$

iv) $\int [f(g(x))] g'(x) dx$

- 1.5. Find the integrals of $\tan x$, $\cot x$, $\sec x$ and $\operatorname{cosec} x$ w.r.t. x .
- 1.6. Evaluate the Standard integrals of the functions of the type

i) $\frac{1}{a^2 + x^2}$, $\frac{1}{a^2 - x^2}$, $\frac{1}{x^2 - a^2}$

ii) $\frac{1}{\sqrt{a^2 + x^2}}$, $\frac{1}{\sqrt{a^2 - x^2}}$, $\frac{1}{\sqrt{x^2 - a^2}}$

iii) $\sqrt{a^2 + x^2}$, $\sqrt{a^2 - x^2}$, $\sqrt{x^2 - a^2}$

- 1.7. Evaluate integrals using decomposition method.
- 1.8. Solve problems using integration by parts.
- 1.9. Use Bernoulli's rule for evaluating the integrals of the form $\int u.v dx$.
- 1.10. Evaluate the integrals of the form $\int e^x [f(x) + f'(x)] dx$
- 1.11. State the fundamental theorem of integral calculus
- 1.12. Explain the concept of definite integral.
- 1.13. Solve simple problems on definite integrals.
- 1.14. State various properties of definite integrals.
- 1.15. Evaluate simple problems on definite integrals using these properties.

Unit -II

Differential Equations

C.O. 2 Obtain differential equations and solve differential equations of first order and first degree.

L.O.2.1. Define a differential equation, its order and degree

- 2.2 Find order and degree of a given differential equation.
- 2.3 Form a differential equation by eliminating arbitrary constants.
- 2.4 Solve the first order and first degree differential equations by variables separable method.
- 2.5 Solve linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x only or constants.

Syllabus for Unit test-I completed

Unit-III

Graph Theory and Probability

C.O. 3 Able to define the basic concepts of Graph Theory and use the principles of Probability in computational systems.

L.O. 3.1 Define a graph.

3.2 Explain the terminology of a graph, vertices, edges, parallel edges, adjacent vertices, self-loops.

3.3 State the significance of Graph Theory in Computer Science applications.

3.4 Explain incidence and degree of a graph.

3.5 Explain the relation between degree and edges of a graph.

3.6 Explain various types of graphs, null graph, trivial graph, simple graph, multigraph, directed graph, non-directed graph and cyclic graph.

3.7 Define walk, path, circuit, length of a graph, distance between two vertices.

3.8 Explain the formation of adjacency matrix of a graph.

3.9 Recall the basic probability principles.

3.10 Define permutations and combinations with examples.

3.11 State addition theorem of probability for two mutually exclusive and exhaustive events.

3.12 Solve simple problems on addition theorem.

3.13 Explain conditional event and conditional probability.

3.14 Solve simple problems on conditional probability.

3.15 Explain dependent, independent events and state multiplication theorem.

3.16 Solve simple problems on multiplication theorem.

3.17 Explain the concept of priori and posteriori probabilities.

3.18 State Bayes' theorem and solve simple problems.

Unit-IV

Statistics

C.O. 4 Apply various statistical techniques for data analysis.

L.O. 4.1 Recall the measures of central tendency.

4.2 Explain the significance of measures of dispersion to determine the degree of heterogeneity of the data.

4.3 Find the measures of dispersion, Range, Mean Deviation and Standard Deviation for ungrouped data.

4.4 Explain the merits and demerits of these measures of dispersion

4.5 Explain bivariate data.

4.6 Explain the concept of covariance and correlation between two variables.

4.7 Calculate Pearson's correlation coefficient between two variables.

4.8 Find Spearman's rank correlation coefficient.

4.9 Explain predictor variables, outcome variables and simple linear regression.

4.10 Calculate the regression coefficients and regression equations with simple problems.

Syllabus for Unit test-II completed

C-23 CM-301

Engineering Mathematics – II

(Common to CM/AIML/AMG/AMT/CAI/CCB/CCN/WD)

CO/PO – Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	2	2	2				3	2	2

CO2	3	2	2	2				3	2	2
CO3	3	3	3	3				3	3	3
CO4	3	3	3	3				3	3	3
Avg.	3	2.5	2.5	2.5				3	2.5	2.5

3 =Strongly mapped (High), **2** = Moderately mapped (Medium), **1** = Slightly mapped (Low)

Note: The gaps in CO/PO mapping will be met with appropriate activities as follows:

- For PO5: Appropriate quiz programmes may be conducted at intervals and duration as decided by concerned faculty.
- For PO6: Seminars on applications of mathematics in various engineering disciplines are to be planned and conducted.
- For PO7: Plan activities in such a way that students can visit the Library to refer standard books on Mathematics and access the latest updates in reputed national and international journals. Additionally, encourage them to attend seminars and learn mathematical software tools.

C-23 CM-301
Engineering Mathematics – II
(Common to CM/AIML/AMG/AMT/CAI/CCB/CCN/WD)
PO - CO – Mapping strength

PO no	Mapped with CO no	CO periods addressing PO in column I		Level (1,2 or 3)	Remarks
		Number	%		
1	CO1, CO2, CO3,CO4	60 (22+8+17+13)	100%	3	>40% Level 3 Highly addressed
2	CO1, CO2, CO3,CO4	38 (8+3+17+10)	63%	3	
3	CO1, CO2, CO3,CO4	38 (8+3+17+10)	63%	3	
4	CO1, CO2, CO3,CO4	38 (8+3+17+10)	63%	3	
5					25% to 40% Level 2 Moderately addressed
6					
7					
PSO 1	CO1, CO2, CO3,CO4	60 (22+8+17+13)	100%	3	5% to 25% Level 1 Low addressed
PSO 2	CO1, CO2, CO3,CO4	38 (8+3+17+10)	63%	3	
PSO 3	CO1, CO2, CO3,CO4	38 (8+3+17+10)	63%	3	

COURSE CONTENTS

Unit-I

Indefinite Integration:

1. Integration regarded as anti-derivative – Indefinite integrals of standard functions - Properties of indefinite integrals - Integration by substitution or change of variable - Integrals of $\tan x$, $\cot x$, $\sec x$, $\operatorname{cosec} x$.

Evaluation of integrals which are of the following forms:

$$i) \frac{1}{a^2 + x^2}, \frac{1}{a^2 - x^2}, \frac{1}{x^2 - a^2}$$

$$ii) \frac{1}{\sqrt{a^2 + x^2}}, \frac{1}{\sqrt{a^2 - x^2}}, \frac{1}{\sqrt{x^2 - a^2}}$$

$$iii) \sqrt{a^2 + x^2}, \sqrt{a^2 - x^2}, \sqrt{x^2 - a^2}$$

Integration by decomposition of the integrand into simple rational, algebraic functions -

Integration by parts, Bernoulli's rule and integrals of the form $\int e^x [f(x) + f'(x)] dx$.

Definite integral-fundamental theorem of integral calculus, properties of definite integrals,
evaluation of simple definite integrals.

Unit -II

Differential Equations:

2. Definition of a differential equation, order and degree of a differential equation, formation of differential equations - Solutions of differential equations of first order and first degree using variables separable method and linear differential equation of the type $\frac{dy}{dx} + Py = Q$.

Unit-III

Graph Theory and Probability

3. Definition of a graph, terminology of a graph, significance in computer science applications - Incidence and degree, relationship between degree and edges - Various types of graphs, null graph, trivial graph, simple graph, multigraph, directed graph, non-directed graph and cyclic graph - Walk, path, circuit, length of a graph, distance between two vertices - Formation of adjacency matrix.
Permutations and Combinations - Addition theorem of probability, conditional probability, dependent and independent events with multiplication theorem - Priori and posteriori probability, Baye's theorem.

Unit III

Statistics

4. Measures of dispersion, range, mean deviation and standard deviation of ungrouped data, merits and demerits - Bivariate data, correlation, Pearson's correlation coefficient, Spearman's rank correlation coefficient - Predictor and outcome variables, simple linear regression coefficients and regression equations.

Textbook:

Reference Books:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
2. Schaum's Outlines Differential Equations, Richard Bronson & Gabriel B. Costa
3. Trembley and Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata-McGraw-Hill.
4. Narsing Deo, Graph Theory, PHI India.
5. Schaum's Outline: Introduction to Probability and Statistics, Seymour Lipschutz & John J. Schiller.

C- 23 Engineering Mathematics – II

Subject Title	:	Engineering Mathematics – II
Subject Code	:	CM/AIML/AMG/AMT/CAI/CCB/CCN/WD-301
Periods/Week	:	04
Periods/Semester	:	60

TIME SCHEDULE

Chapter. No	Chapter/Unit title	No. of Periods	Marks Allotted	Short Type	Essay Type	COs mapped
Unit – I: Integral Calculus						
1	Indefinite integration	17	26	2	2	CO1
2	Definite integrals	5	16	2	1	CO1
Unit – II: Differential Equations						
3	Introduction to Differential equations	2	3	1	0	CO2
4	Solutions of Differential equations of first order	6	10	0	1	CO2
Unit – III: Graph Theory and Probability						
5	Graph theory	5	6	2	0	CO3
6	Probability	12	26	2	2	CO3
Unit – IV : Statistics						
6	Measures of Central Tendency	1	0	0	0	CO4
7	Measures of Dispersion	3	3	1	0	CO4
8	Correlation	4	10	0	1	CO4
9	Simple linear regression	5	10	0	1	CO4
Total		60	110	10	8	
Marks				30	80	

C-23 CM-301
Engineering Mathematics – II
(Common to CM/AI ML/AMG/AMT/CAI/CCB/CCN/WD)

Unit Test Syllabus

Unit Test	Syllabus
Unit Test-I	From L.O 1.1 to L.O 2.5
Unit Test-II	From L.O 3.1 to L.O 4.10

Course Code	Course Title	No. of Periods/ Weeks	Total No. of periods	Marks for FA	Marks for SA
CM-302	Digital Electronics	5	75	20	80

Time Schedule

Chapter No.	Chapter/Unit Title	No. of Periods	Marks	No. of Short Answer Questions	No. of Essay Type Questions	CO's Mapped
1.	Number systems	15	16	2	1	CO1
2.	Logic Gates ,Boolean Algebra and basic Combinational circuits	17	26	2	2	CO2
3.	Flip-Flops	15	26	2	2	CO3
4.	Counters and registers	17	26	2	2	CO4
5.	Additional Combinational circuits	11	16	2	1	CO5
Total		75	110	10	8	

Course Objectives	Upon completion of the course the student shall be able
	i)To acquire the basic knowledge of digital logic levels and apply of knowledge to understand digital logic circuits. ii)To prepare students to perform the analysis and design of various digital electronic circuits.

Course Outcomes

Upon completion of the course the student shall be able to		
CO1	CM-302.1	Explain the structure of various number systems.
CO2	CM-302.2	Describe fundamental concepts and techniques used in digital electronics, the switching algebra theorems and logic gates and apply them to design logic circuits using K-Maps
CO3	CM-302.3	Explain operation of flip flops
CO4	CM-302.4	Design the counting circuits, Registers using flip flop operations.
CO5	CM-302.5	Explain design and implement various sequential circuits, combinational circuits and PLD

CO-PO/PSO Matrix:

CO NO.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CM-302.1	3	3	3		2			3	2	3
CM-302.2	3	3	3	2	1	1		3	2	2
CM-302.3	2	1	1	1			1	1	1	3
CM-302.4	2	2	3	1		2	1	2	2	1
CM-302.5	2	2	3	1	1	1	2	3		1
Average	2.4	2.2	2.6	2.25	1.3	1.3	1.3	2.4	1.75	2

3=Strongly mapped , 2=moderately mapped, 1=slightly mapped

Learning Outcomes:

1.0 Number systems

1.1 List the various number systems used in digital Computer.

1.2 Explain Decimal number system

1.3 Explain Binary number system

1.4 Explain octal number system

1.5 Explain Hexadecimal number system

1.6 Convert decimal number to other base conversion.

1.6.1 Decimal to Binary

1.6.2 Decimal to Octal

1.6.3 Decimal to Hexadecimal

1.7 Convert binary number to other base conversion.

1.7.1 Binary to Decimal

1.7.2 Binary to octal

1.7.3 Binary to Hexadecimal

1.8 Convert octal number to other base conversion.

1.8.1 Octal to Decimal

1.8.2 Octal to Binary

1.8.3 Octal to Hexadecimal

1.9 Convert hexadecimal other base conversion.

1.9.1 Hexadecimal to Decimal

1.9.2 Hexadecimal to Binary

1.9.3 Hexadecimal to Octal

1.10 Binary numbers representation.

1.10.1 Define Binary numbers representation.

1.10.2 List the types of Binary numbers representation.

1.10.3 Explain Unsigned binary number representation.

1.10.4 Explain Signed binary number representation.

1.11 Signed binary arithmetic.

1.11.1 Illustrate addition of two signed binary numbers.

- 1.11.2 Illustrate subtraction of two signed binary numbers.
- 1.11.3 Illustrate binary multiplication.
- 1.11.4 Illustrate Binary division.
- 1.12 Binary coded decimal (BCD) coding scheme.**
 - 1.12.1 Define Binary coded decimal (BCD) coding scheme.
 - 1.12.2 List the types of Binary coded decimal (BCD)
 - 1.12.3 Draw and explain 8421 code.
 - 1.12.4 Draw and explain 2421 code.
 - 1.12.5 Draw and explain 8 4-2-1 code.
 - 1.12.6 Draw and explain Excess 3 code.
- 1.13 Character representation**
 - 1.13.1 List character representation codes
 - 1.13.2 Explain the ASCII coding scheme.
 - 1.13.3 Explain the EBCDIC coding scheme.
- 2.0 Boolean algebra , Logic gates and Basic Combinational Circuits**
- 2.1 Boolean algebra**
 - 2.1.1 Define Boolean algebra
 - 2.1.2 Explain AND, OR, NOT operations with truth tables.
 - 2.1.3 Explain the working of EX-OR and EX-NOR operations with truth tables.
 - 2.1.4 List the different postulates in Boolean algebra.
 - 2.1.5 State De-Morgan's theorems.
 - 2.1.6 Prove De-Morgan's theorems using truth tables.
 - 2.1.7 Apply De-Morgan's theorems and other postulates of Boolean algebra to simplify the given Boolean expression.
 - 2.1.8 Generate Boolean expression for given truth table.
 - 2.1.9 Using Sum-Of-Products(SOP) method
 - 2.1.10 Using Product-Of-Sums(POS)method
 - 2.1.11 Use K – map to simplify Boolean expression (up to 4 variables).
 - 2.1.12 Using Two variable K-Map
 - 2.1.13 Using Three variable K-Map
 - 2.1.14 Using Four variable K-Map
- 2.2 Logic Gates**
 - 2.2.1 Define Logic gate
 - 2.2.2 List basic gates
 - 2.2.3 Define OR gate
 - 2.2.4 Explain OR gate with logic symbol and truth table.
 - 2.2.5 Define AND gate
 - 2.2.6 Explain AND gate with logic symbol and truth table.
 - 2.2.7 Define NOT gate
 - 2.2.8 Explain NOT gate with logic symbol and truth table.
 - 2.2.9 What is universal gate? List universal gates
 - 2.2.10 Define NOR gate
 - 2.2.11 Explain NOR gate with logic symbol and truth table.
 - 2.2.12 Define NAND gate
 - 2.2.13 Explain NAND gate with logic symbol and truth table.
 - 2.2.14 Define EX-OR and EX-NOR gates
 - 2.2.15 Explain the working of EX-OR and EX-NOR gates with truth tables.
 - 2.2.16 Implement AND, OR, NOT gates using NAND gates only
 - 2.2.17 Implement AND, OR, NOT gates using NOR gate only.
- 2.3 Basic Combinational Circuits**
 - 2.3.1 Define the Half Adder.
 - 2.3.2 Explain the function of Half Adder.
 - 2.3.3 Draw Half-Adder circuit using an exclusive OR and an AND gate.
 - 2.3.4 Draw a Half-Adder using only NAND gates or only NOR gates.

- 2.3.5 Define the Full Adder.
- 2.3.6 Explain the function of Full Adder.
- 2.3.7 Construct Full Adder using two Half-Adder and an OR gate
- 2.3.8 Define the parallel Adder
- 2.3.9 Explain the function of parallel Adder using logic symbol.
- 2.3.10 Draw and explain 4-bit parallel adder using full adders.
- 2.3.11 Draw and explain 4-bit parallel adder/ 2's complement subtractor circuit.
- 2.3.12 Explain the working of a serial adder with a block diagram.
- 2.3.13 List the advantage and disadvantages of a serial adder
- 2.3.14 List the advantage and disadvantages of a parallel adder.
- 2.3.15 Distinguish between serial adder and parallel adder.
- 2.3.16 Explain the operation of a digital comparator circuit for two 4-bit words.
- 2.4 Flip-flops**
- 2.5 List the details of different logic families.
- 2.6 Define positive and negative logic levels.
- 2.7 Define Flip flop
- 2.8 Draw and explain the basic principle of operation of a Flip-flop.
- 2.9 Define Latch.
- 2.10 Explain the working of a NAND latch circuit with truth table and Timing diagram
- 2.11 Explain the working of a NOR latch circuit with truth table and Timing diagram
- 2.12 Differentiate between Latch and Flip-flop.
- 2.13 Define Triggering
- 2.14 List the types of Triggering
- 2.15 Draw and explain the concept of edge triggering(positive, negative)
- 2.16 Draw and explain the concept of level triggering. (positive, negative)
- 2.17 Explain with block diagram, waveforms and truth tables the working of RS Flip-flop.
- 2.18 Explain with block diagram, waveforms and truth tables the working of RS T Flip-flop.
- 2.19 Explain with block diagram, waveforms and truth tables the working of D Flip-flop.
- 2.20 Explain with block diagram, waveforms and truth tables the working of JK Flip-flop.
- 2.21 Explain with block diagram, waveforms and truth tables the working of T Flip-flop.
- 2.22 Distinguish between synchronous and asynchronous inputs of a flip- flop
- 2.23 State the need for a Master-Slave flip-flop.
- 2.24 Explain the working of a Master-Slave flip-flop using suitable circuit diagram and truth table.
- 3.0 Counters and Registers**
- 3.1 Counters**
- 3.1.1 Define Counter
- 3.1.2 List the types of counters.
- 3.1.3 Define terms Synchronous counter, Asynchronous counter
- 3.1.4 Distinguish between asynchronous and synchronous counters.
- 3.1.5 Draw and explain module-8 ripple counter circuit diagram with waveforms and truth tables
- 3.1.6 Draw and explain module-16 ripple counter circuit diagram with waveforms and truth tables
- 3.1.7 Draw and explain module-10 (decade) Asynchronous counter circuit diagram with waveforms and truth tables
- 3.1.8 Draw and explain module-8 synchronous counter circuit diagram with waveforms and truth tables
- 3.1.9 Draw and explain module-16 synchronous counter circuit diagram with waveforms and truth tables
- 3.1.10 List the drawbacks of ripple counters.
- 3.1.11 List the advantages of synchronous counters
- 3.1.12 Explain the operation of a up/down counter circuit diagram with waveforms and truth tables

- 3.1.13 State the need of Programmable counter
- 3.1.14 Draw and explain Programmable counter
- 3.1.15 Draw and explain the operation of a 4-bit ring counter.
- 3.1.16 List the applications of counter.
- 3.2 Register**
 - 3.2.1 Define Register
 - 3.2.2 State the need of Register.
 - 3.2.3 List the methods of data transfer in register.
 - 3.2.4 List the types of Registers
 - 3.2.5 Define Serial in – Serial out register
 - 3.2.6 Define Serial in – Parallel out register
 - 3.2.7 Define Parallel in – Serial out register
 - 3.2.8 Define Parallel in – Parallel out register
 - 3.2.9 Explain the working of serial in – serial out register with circuit diagram.
 - 3.2.10 Explain the working of serial in – parallel out register with circuit diagram.
 - 3.2.11 Explain the working of shift left Register with circuit diagram.
 - 3.2.12 Explain the working of shift right registers with circuit diagram.
 - 3.2.13 Explain the working of universal shift register.
 - 3.2.14 Draw and explain the use of shift register as memory.
- 4.0 Additional Combinational Circuits**
 - 4.1 Define data selector and state its importance.
 - 4.2 Define data distributor and state its importance.
 - 4.3 Define the terms Multiplexer and De multiplexer
 - 4.4 Draw and explain the operation of a Multiplexer circuit diagram with truth table.
 - 4.5 Draw and explain the operation of De multiplexer circuit diagram with truth table.
 - 4.6 List the applications of Multiplexers.
 - 4.7 List the applications of De multiplexers.
 - 4.8 Define the terms Encoder and Decoder.
 - 4.9 Draw and explain the operation of a 4 to 10 line decoder circuit diagram with truth table.
 - 4.10 Draw and explain the operation of a 8 to 3 encoder circuit diagram with truth table.
 - 4.11 List applications of Decoders.
 - 4.12 List applications of Encoders.
 - 4.13 Define Programmable logic Devices.
 - 4.14 List the types of Programmable logic Devices.
 - 4.15 Draw and explain the Programmable Logic Array(PLA).

COURSE CONTENTS

- 1. Number Systems:** Number Systems, Decimal, Binary, Hexadecimal and Octal codes, Conversion from one number system to another number system, Binary numbers representation ,Binary Arithmetic, BCD ,Character representation-ASCII and EBCDIC code for characters.
- 2. Boolean algebra and Logical Gates :**AND, OR, NAND, NOT, NOR & EX-OR gates. Logical definitions – Symbols – truth tables. Boolean theorems, Boolean simplifications of Boolean expressions, Using De-Morgan's theorems, Formation and implementation of Logic expressions, Karnaugh's mapping, Applications involving developing of combinational logic circuits. Half-Adder, Full-adder, Subtractor series – Parallel Binary adder – Parallel adder/sub tractor circuits.
- 3. FLIP FLOP: Different logic families,** Basic principles of Flip Flop operation (with help of wave form & truth tables) of RS,T,D,JK and Master Slave JK flip flop concept of Edge Triggering and Level Triggering , Synchronous and Asynchronous device.
- 4. Counters:** Basic Asynchronous, Synchronous Binary and Decade counter and the Ripple counter, their use Decade counter, Up and Down counters, Ring counter

Registers : Shift registers, Serial, Parallel register, Serial-in Parallel out, Parallel-in–serial out devices, Universal shift registers, Applications.

5. **Additional Combinational Circuits:** Multiplexers, De multiplexers and Encoders, Decoders-operation of a multiplexer with a circuit diagram - operation of a de multiplexer with a circuit diagram - applications of multiplexers and de multiplexers-operation of a 4 to 10 line decoder - operation of an 8 to 3 line encoder – programmable logic array.

REFERENCE:

1. Digital principles and applications ... Malvino and Leach
2. Digital Electronics.... Bignell – Thomson
3. Modern Digital Electronics.... R.P. JAIN
4. <https://www.javatpoint.com/number-system-in-digital-electronics>
5. <https://www.electronics-tutorials.ws/binary/binary-coded-decimal.html>
6. <https://www.javatpoint.com/excess-3-code-in-digital-electronics>
7. <https://www.tutorialspoint.com/conversion-of-binary-to-gray-code>
8. <https://atozmath.com/example/NumToBaseConv.aspx?he=e&b1=111&b2=2>

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.7
Unit test-2	From 3.8 to 5.7

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CM-303	Operating Systems	4	60	20	80

Time Schedule

Chapter No.	Chapter/Unit Title	No. of Periods	Marks	No. of Short Answer Questions	No. of Essay Type Questions	CO's Mapped
1.	Introduction to Operating system	12	16	2	1	CO1
2.	Process management	12	16	2	1	CO2
3.	Synchronization & Deadlocks	12	26	2	2	CO3
4.	Memory management	12	26	2	2	CO4
5.	Disk scheduling and File management	12	26	2	2	CO5
Total		60	110	10	8	

Course Objectives	i)To know about the basics of Operating Systems ii)To familiarize with process management, Scheduling algorithms, Synchronization and deadlock techniques iii)To understand various Memory management techniques iv)To familiarize with File management
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Course Out comes	CO1	CM-303.1	Explain basic concepts of Operating System
	CO2	CM-303.2	Explain process scheduling algorithm
	CO3	CM-303.3	Describe Semaphores, synchronization and Deadlock handling techniques
	CO4	CM-303.4	Use memory management techniques and page replacement algorithms

	CO5	CM-303.5	Use Disk scheduling algorithms and File allocation methods with respect to different operating systems
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CO-PO/PSO MATRIX

CO NO.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CM-303.1	3	2	2	2	3	3	3	3	2	2
CM-303.2	3	3	3	2	2	3	3	2	2	2
CM-303.3	3	3	3	2	3	3	3	3	2	3
CM-303.4	3	3	3	3	3	3	3	3	3	3
CM-303.5	3	3	2	2	2	3	3	3	3	3
Average	3	3	2.5	2	2.5	3	3	3	2.5	2.5

3=Strongly mapped , 2=moderately mapped, 1=slightly mapped

Learning Outcomes:

1.0 Introduction to operating systems

- 1.1 Define an operating system
- 1.2 Discuss history of operating system
- 1.3 Discuss about various types of operating systems
- 1.4 Distinguish spooling and buffering
- 1.5 Explain the concepts multiprogramming and timesharing
- 1.6 Differentiate between distributed and real time systems
- 1.7 Describe multiprocessor systems
- 1.8 Describe the operating system components
- 1.9 Discuss operating system services
- 1.10 Define system call with an example
- 1.11 List and explain different types of system calls
- 1.12 Define single user, multi user operating system structure

2.0 Process management

- 2.1 Define process and process control block
- 2.2 Explain process state diagram
- 2.3 Describe process creation and termination
- 2.4 Discuss the relation between processes
- 2.5 Define Thread and describe multithreading
- 2.6 Explain scheduling concepts
- 2.7 Describe scheduling queues and schedulers
- 2.8 Explain CPU scheduling and scheduling criteria
- 2.9 Explain various scheduling algorithms
 - 2.9.1 FCFS
 - 2.9.2 SJF
 - 2.9.3 Round Robin

3.0 Synchronization & Deadlocks

- 3.1 Describe semaphores
- 3.2 Explain inter process communication
- 3.3 Define Deadlock
- 3.4 State the necessary conditions for arising deadlocks
- 3.5 State various techniques for deadlock prevention
- 3.6 Discuss Deadlock avoidance and detection
- 3.7 Describe the process of recovering from deadlock

4.0 Memory management

- 4.1 Describe briefly address binding, dynamic loading, dynamic linking
 - 4.2 Define overlays
 - 4.3 Describe briefly on swapping
 - 4.4 Explain single partition allocation
 - 4.5 Explain multiple partition allocation
 - 4.6 Explain the concept of fragmentation
 - 4.7 Explain paging concept
 - 4.8 Explain how logical address is translated into physical address
 - 4.9 Explain segmentation and segmentation with paging
 - 4.10 Define virtual memory techniques
 - 4.11 Describe demand paging
 - 4.12 Describe page replacement
 - 4.13 Discuss on page replacement algorithms
 - 4.13.1 FIFO
 - 4.13.2 LRU
 - 4.13.3 Optimal
 - 4.14 Explain the concept of thrashing
 - 4.15 Explain working set model and page fault frequency
- 5.0 Disk scheduling and File management**
- 5.1 List out various disk performance parameters
 - 5.2 Disk scheduling policies
 - 5.2.1 FIFO
 - 5.2.2 SSTF
 - 5.2.3 SCAN
 - 5.3 Define file management
 - 5.4 List and explain various file operations
 - 5.5 List and explain various access methods
 - 5.6 List and explain various allocation methods
 - 5.6 List and explain directory structure
 - 5.7 Explain disk organization and structure

COURSE CONTENT

1.0 Introduction to operating systems

Operating System –Evolution of operating system-Types of Operating Systems - Multi Programming and Time Sharing - Distributed and Real time Systems - spooling and buffering - Multi processor systems-Components of Operating Systems - operating System Services - system Calls - single User and Multi user operating System Structure.

2. Process management

Processes - Sequential Processes - Process State Diagram - Process Control Block - Process Creation and Termination - Relations between Processes - Threads and Multi Threading - Scheduling Concepts • Schedulers - CPU scheduling and Scheduling criteria - scheduling algorithms.

3. Synchronization & Deadlocks

Inter Process Communications - semaphores – monitors

Deadlocks - principal of deadlock - deadlock prevention - deadlock detection - deadlock avoidance.

4. Memory management

Address binding -Dynamic Loading- dynamic linking-overlays-swapping• memory allocation-fragmentation-paging-segmentation- segmentation with paging-

Benefits of virtual memory - virtual memory techniques - demand paging - page replacements - page replacement algorithms – thrashing.

5. Disk scheduling and File management

Disk performance parameters - Disk scheduling policies – Introduction to file systems - File management-File Operations - Access methods - Directory structure organization - File Protection.

REFERENCE BOOKS

- | | | |
|-------------------------------|----|--------------------------|
| 1. Operating Systems | -- | Silber Schatz and Galvin |
| 2. Operating Systems | -- | William Stallings, PHI |
| 3. Operating Systems | -- | Dietel and Dietel |
| 4. Operating Systems | -- | Dhamdhare (TMH) |
| 5. Advanced Operating Systems | -- | Tanenbaum |

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.2
Unit test-2	From 3.3 to 5.7

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CM-304	Data Structures Through C	5	75	20	80

Time Schedule

Chapter No.	Chapter/Unit Title	No. of Periods	Marks	No. of Short Answer Questions	No. of Essay Type Questions	CO's Mapped
1.	Introduction to Data Structures Sequential Storage Representation	23	16	2	1	CO1
2.	Linked Storage Representation-Linked Lists	15	16	2	1	CO2
3.	Linear Data Structures-Stacks	12	26	2	2	CO3
4.	Linear Data Structures-Queues	10	26	2	2	CO4
5.	Non Linear Data Structures-Trees	15	26	2	2	CO5
Total		75	110	10	8	

Course Objectives	At the end of the course students will be able to
	i)To know the various types of Data Structures ii)To familiarize with the representation of Data Structures iii)To use various Data structures in organizing data iv)To reinforce theoretical concepts by writing relevant programs

COURSE OUTCOMES:

CO1	CM-304.1	Illustrate various techniques of sorting and searching
CO2	CM-304.2	Explain the operations on Various Linked Lists
CO3	CM-304.3	Apply the operations of Stack.
CO4	CM-304.4	Explain the operations of different types of Queue.

CO5	CM-304.5	Apply Binary tree traversal techniques.
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CO-PO/PSO MATRIX

CO NO.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CM-304.1	3	3	3	1	3	3	3	3	2	2
CM-304.2	3	3	3	1	3	3	3	2	3	2
CM-304.3	3	3	3	2	3	3	3	2	2	2
CM-304.4	3	3	3	2	3	3	3	2	2	2
CM-304.5	3	3	3	2	3	3	3	2	3	3
Average	3	3	3	2	3	3	3	2	2.5	3

3=Strongly mapped , 2=moderately mapped, 1=slightly mapped

Learning Outcomes:

1. Introduction to Data Structures

1.1 Understand various types of Data Structures

- 1.1.1. Define Data Structure and classify them
- 1.1.2. Explain Linear Data Structures
- 1.1.3. Describe Non-Linear Data Structures
- 1.1.4. Explain Data Types and Abstract Data Types
- 1.1.5. Explain about Space and Time Complexities

1.2 Sequential Storage Representation

- 1.2.1 Various Sorting Techniques
 - 1.2.1.1 Define Sorting
 - 1.2.1.2 State the need of Sorting
 - 1.2.1.3 List the methods of Sorting
 - 1.2.1.4 Explain the following for Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort
 - 1.2.1.4.1 Sorting technique,
 - 1.2.1.4.2 Algorithm and complexity
 - 1.2.1.4.3 Program
- 1.2.2 Various Searching Techniques
 - 1.2.2.1 Define searching
 - 1.2.2.2 State the need of searching
 - 1.2.2.3 List two types of searching
 - 1.2.2.4 Explain the following for Linear Search, Binary Search with Recursion and with-out Recursion
 - 1.2.2.4.1 Searching technique.
 - 1.2.2.4.2 Algorithm and complexity
 - 1.2.2.4.3 Program

2. Linked Storage Representation –Linked Lists

- 2.1 List the advantages & disadvantages of Linked Lists over Arrays
- 2.2 State the purpose of Dummy Header
- 2.3 Explain the following for Singly Linked List, Doubly Linked List
 - 2.3.1. Structure
 - 2.3.2. Creation
 - 2.3.3. How to represent a node using 'C' Structure
 - 2.3.4. Perform insertion, deletion, traverse and sort operations
 - 2.3.5. Perform search and replace an element
 - 2.3.6. C Program with all operations.
- 2.4 Singly circular list
 - 2.4.1 Definition
 - 2.4.2 Structure
 - 2.4.3 Advantages of a Singly Circular Linked List over a Singly Linked List
 - 2.4.4 The changes require in a singly linked list program to make it the Singly Circular List.
- 2.5 Doubly circular list
 - 2.5.1 Definition
 - 2.5.2 Structurer
 - 2.5.3 Advantages of a Doubly Circular Linked List over a Doubly Linked List
 - 2.5.4 The changes require in a Doubly Linked List program to make it the Doubly Circular List.

3. Linear Data Structures-Stacks

- 3.1 Define Stack
- 3.2 Explain the push, pop and display operations of a Stack
- 3.3 Explain array implementation of a Stack with various operations.
- 3.4 Explain the program for Array implementation of a Stack with various operations.
- 3.5 Explain Linked List implementation of a Stack with various operations.
- 3.6 Explain the program for Linked List implementation of a Stack with various operations.
- 3.7 List the applications of Stacks
- 3.8 Convert Infix expression to Postfix expression
- 3.9 Explain the program for Conversion of Infix expression to Postfix expression
- 3.10 Evaluate Postfix expression
- 3.11 Explain the program for Evaluating Post fix expression

4. Linear Data Structures-Queues

- 4.1 Define Queue
- 4.2 Explain the insertion, deletion and display operations on Queues
- 4.3 Explain array implementation of a Queue with various operations.
- 4.4 Explain the program for Array implementation of a Queue with various operations.
- 4.5 Explain Linked List implementation of a Queue with various operations.
- 4.6 Explain the program for Linked List implementation of a Queue with various operations.
- 4.7 Know about Circular Queues
- 4.8 Explain array implementation of a Circular Queue with various operations
- 4.9 Explain the program for Array implementation of a Circular Queue with various operations
- 4.10 Explain Linked List implementation of a Circular Queue with various operations.

- 4.11 Explain the program for Linked List implementation of a Circular Queue with various operations.
- 4.12 List the application of Queues
- 4.13 Know about Priority Queues

5. Non Linear Data Structures-Trees

- 5.1 Define a Tree
- 5.2 State importance of various terminologies related to Tree:
 - 5.2.1 Root, Edge, Parent, Child, Siblings, Leaf, Internal nodes, Degree, Level, Height, Depth, Path, Sub tree, Forest.
- 5.3 Define Binary Tree
- 5.4 Differences between General Tree and Binary Tree.
- 5.5 Conversion of General Trees to Binary Trees
- 5.6 Explain the linear representation and linked list representation of a Binary Tree
- 5.7 Define Binary Search Tree
- 5.8 Differences between Binary Search Tree and Binary Tree
- 5.9 Perform various traversals on Binary Search Trees
- 5.10 Construct a Binary Tree using In-order and Preorder Traversals
- 5.11 Construct a Binary Tree using In-order and Post-order Traversals
- 5.12 Know the importance of Binary Search Trees over General Trees
- 5.13 Perform insertion, deletion, search and various traversal operations on a Binary Search Tree.
- 5.14 Explain the program of Binary Search Tree with all operations.
- 5.15 List the Applications of trees

COURSE CONTENT

1. Introduction to Data Structures

Data structures – Linear & non linear, data types and abstract data types, algorithm analysis for time and space requirements.

Sequential Storage Representation – Sorting - Introduction to different sorting techniques – Bubble, Selection, Insertion, Quick & Merge. **Searching** – Introduction to different searching techniques – Linear and Binary.

2. Linear data structures-Linked Lists

Linked Lists – Types - Singly Linked Lists – Create, insert, delete, sort, search and replace an element in a linked list – - Doubly Linked Lists – Create, insert, delete, sort, search and replace an element in a linked list – Define terms Singly circular and doubly circular linked lists

3. Linear data structures-Stacks

Stacks- Implementation of stacks, application of stacks, converting infix to postfix expression and postfix expression evaluation.

4. Linear data structures-Queues

Queues–Implementation of queues- Application of queues- know about Circular queues, and Priority queue.

5. Non Linear data structures-Trees

Trees- Trees- Trees Terminology–Binary trees –Representation – Linear and Linked list representation-Binary Search Tree-various operations-Tree traversals-Tree Conversions& Applications

REFERENCE BOOKS

- 1. Data Structures: A Pseudocode Approach with C - Gilberg / Forouzan

2. Data Structures using 'C' - TanenbaumLangsam and Augenstein(PHI).
3. Data structures through C - Yashwanth Kanetkar
4. An Introduction to data structures with applications - Tremblay & Sorenson

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 2.7
Unit test-2	From 3.1 to 5.15

Course code	Course Title	No. of Periods/Weeks	Total No. of periods	Marks for FA	Marks for SA
CM-305	DBMS	5	75	20	80

Time Schedule

Chapter No.	Chapter/Unit Title	No. of Periods	Marks	No. of Short Answer Questions	No. of Essay Type Questions	CO's Mapped
1.	Concepts of DBMS & RDBMS	18	29	3	2	CO1
2.	Concepts of SQL	22	26	2	2	CO2
3.	Basics of PL/ SQL	15	26	2	2	CO3
4.	Advance PL/SQL	10	16	2	1	CO4
5.	Concepts of No SQL & Mongo DB.	10	13	1	1	CO5
Total		75	110	10	8	

Course Objectives	i)To know the fundamentals of DBMS ii)To familiarize insert, retrieve, update, delete data in database iii)To familiarize programming skills for insert, retrieve, update, delete data in database
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Course outcome:

CO1	CM-305.1	Describe fundamentals, types and Overall structure of DBMS
CO2	CM-305.2	Apply SQL commands to create, retrieve, update, delete data from the Relational data bases.
CO3	CM-305.3	Describe PL/SQL programming constructs, control statements and sub programs.
CO4	CM-305.4	Apply cursors, triggers and Exception handling concepts
CO5	CM-305.5	Use NOSQL database concepts and MongoDB data base concepts in designing database Schema.

CO-PO/PSO MATRIX

CO NO.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CM-305.1	3	2	2	2	2	3	2	2	3	1
CM-305.2	2	3	3	3	3	3	3	2	1	2
CM-305.3	3	2	2	2	1	3	2	2	3	1
CM-305.4	1	3	2	3	2	2	3	2	3	3
CM-305.5	3	1	3	1	3	3	2	2	2	3
Average	2.4	2.2	2.6	2	2.2	3	2.8	2	2	2

3=Strongly mapped , 2=moderately mapped, 1=slightly mapped

Learning Outcomes:

1.0 Concepts of DBMS & RDBMS

- 1.1** Define Database Management System(DBMS)
- 1.2** List the advantages of DBMS
- 1.3** Explain Database Abstraction, Data Independence
- 1.4** Define Instances and schemas
- 1.5** Explain Data Models.
- 1.6** Define Database languages DDL, DML, TCL
- 1.7** Explain Database Administrator, Users and Database System Architecture with diagram.
- 1.8** Define Entity, Entity sets, Relationship, Relationship sets, Super Key , Candidate Key and Primary Key, Foreign Key
- 1.9** Explain Mapping Cardinalities.
- 1.10** List the symbols used in ER model.
- 1.11** Know The Entity-Relationship Model.
- 1.12** Reduce the ER-diagrams to tables
- 1.13** Explain Generalization, Specialization & Aggregation.
- 1.14** Explain Functional Dependencies, Normalizations– 1 NF, 2 NF and 3NF

2.0 Concepts of SQL

- 2.1** Explain SQL and benefits of SQL.
- 2.2** Describe about Embedded SQL and Lexical conventions
- 2.3** Describe Naming of the Objects and parts and how to refer them.
- 2.4** Explain literals & different data types like character, number, long, date, raw and long raw etc.
- 2.5** Illustrate the comments within SQL Statement
- 2.6** Explain SQL Operators
- 2.7** Describe Data Definition Language commands CREATE, ALTER and DROP.
- 2.8** Explain integrity constraints through creating a table and altering table.
- 2.9** Describe Data Manipulation Language commands INSERT, UPDATE and DELETE
- 2.10** Explain SELECT statement with WHERE, ORDER BY, GROUP BY and HAVING clauses with examples
- 2.11** List and explain single row (Number, character, date and conversion) functions
- 2.12** List and Explain group functions
- 2.13** Explain Transaction Control Commands COMMIT, SAVEPOINT, ROLLBACK, GRANT, and REVOKE.
- 2.14** Explain Sub queries with examples
- 2.15** Explain Joins (Equi Join, Non-Equi Joins, Inner Join, Outer Join, cross join and Self join) with syntax and examples.

3.0 PL/SQL

- 3.1** Explain PL/SQL Block structure.
- 3.2** List the features of PL/SQL
- 3.3** Explain the data types of PL/SQL
- 3.4** Declaration of variables
- 3.5** Explain PL/SQL tables and user defined records.
- 3.6** Explain Input / Output statements
- 3.7** Explain decision making statements and illustrate
- 3.8** Explain looping statements and illustrate
- 3.9** Define procedure and function
- 3.10** Describe the advantages of subprograms.
- 3.11** Explain handling procedures and functions with example programs.
- 3.12** Explain the parameter modes in PL/SQL with examples (in , out and in out)

4.0 Advanced PL/SQL

- 4.1** Define cursor.
- 4.2** Classify cursors
- 4.3** Explain implicit cursor with example
- 4.4** Explain explicit cursors with example
- 4.5** Define trigger
- 4.6** List Advantages of triggers
- 4.7** Explain database triggers.

5.0 Concepts of No SQL & Mongo DB.

5.1 No SQL

- 5.1.1 List features of NOSQL
- 5.1.2 Compare RDBMS and No SQL
- 5.1.3 List the Advantages and Disadvantages of No SQL
- 5.1.4 Know about the ACID and BASE system.
- 5.1.5 Compare ACID and BASE properties
- 5.1.6 No SQL
 - 5.1.6.1 Key-value stores,
 - 5.1.6.2 Column-oriented,
 - 5.1.6.3 Graph oriented Databases
 - 5.1.6.4 Document oriented Databases.

5.2 Mongo DB

- 5.2.1 What is Mongo DB
- 5.2.2 List the advantages of Mongo DB
- 5.2.3 Explain the Creation, Dropping, Creation of Collection
- 5.2.4 Dropping of Collection of Database in Mongo DB
- 5.2.5 Explain the Data types of Mongo DB.
- 5.2.6 Explain Inserting Document, Query Document, Update Document, Deleting Document & Sorting Document.

COURSE CONTENT

1. Concepts of DBMS & RDBMS

Define DBMS –Purpose of DBMS - Data Abstraction – Data Models – Instances and Schemas – Data Independence – Data Definition Language - Data Manipulation Language – Database Administrator - Database Users – Database system Structure.
Entities – Relationships and Relationship sets – Mapping constraints – Entity – Relationship Diagram – Super key , Candidate key and Primary key - Reducing E- R Diagrams to tables – Generalization and Specialization – Aggregation – Functional Dependencies - Normal forms 1NF , 2 NF , 3 NF

2. Concepts of SQL

Benefits of SQL – Embedded SQL – Lexical conventions – Naming objects and parts – Referring objects and parts – Literals – Text –Integer – Number – Data types – Character data types – Number data type – Long data type –Raw and Long Raw data types –Pseudo columns – comments within SQL statements – comments on schema objects.
Operators – Unary and Binary operators – Precedence- Arithmetic operators – character operators – comparison operators – logical operators- set operators – other operators – DDL Commands – Integrity Constraints – DML Commands - functions – single row functions – numeric functions – character functions – date functions – conversion functions – other functions- Group functions. Transaction control commands-Sub queries - Joins.

3. Basics of PL/SQL

Main features – architecture – advantage of PL/SQL – fundamentals – character set – Lexical units – Data types – data type conversion – Declaration - scope and visibility – assignments – expressions and comparisons – PL/SQL tables – user defined records. Conditional control- IF statement – sequential control- GOTO and NULL statements. SQL support – national language support – Remote Access
Advantages of subprograms – procedures – Functions RETURN statement – forward declarations – actual versus formal parameters – positional and named notation - parameter modes

4. Advanced PL/SQL

Cursors – Implicit cursor – Explicit cursor – Triggers – Advantages - creating trigger – raising trigger -

5. No SQL & Basics of Mongo DB

Classification of Databases : RDBMS, OLAP, No SQL-Introduction to No SQL- need for No SQL – Comparison of RDBMS and No SQL- Advantages and Disadvantages of No SQL - BASE system – ACID System – Comparison of ACID and BASE properties – Classification of No SQL as Key-value stores, Column-oriented, Graph and Document oriented Databases
Introduction to Mongo DB - advantages of Mongo DB - applications of Mongo DB - Installation of Mongo DB - Creation of Database - Dropping of Database - Creation of Collection - Dropping of Collection - Data types of Mongo DB - different Commands of Mongo DB - Inserting Document - Query Document - Updating Document – Deleting Documents - Sorting Documents

REFERENCE BOOKS

1. Database System Concepts -- Silber schatz, Henry F. Korth, S. Sudarshan
2. Oracle Database 11g :The Complete Reference - Kevin Loney
3. Understanding ORACLE -- James T. Peary & Joseph G. Laseer.
4. RDBMS with ORACLE -- Rolland.
5. ORACLE series books of ORACLE Press – TMH.
6. Starting out with Oracle – Covering Databases- John Day & Craig Van
7. PL/SQL, Developer Tools & DBA -- Slyke, Dreamtech
8. www.nosql-database.org
9. www.mongodb.org
10. <https://www.guru99.com/nosql-tutorial.html#5>

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 3.5
Unit test-2	From 3.6 to 5.2.5